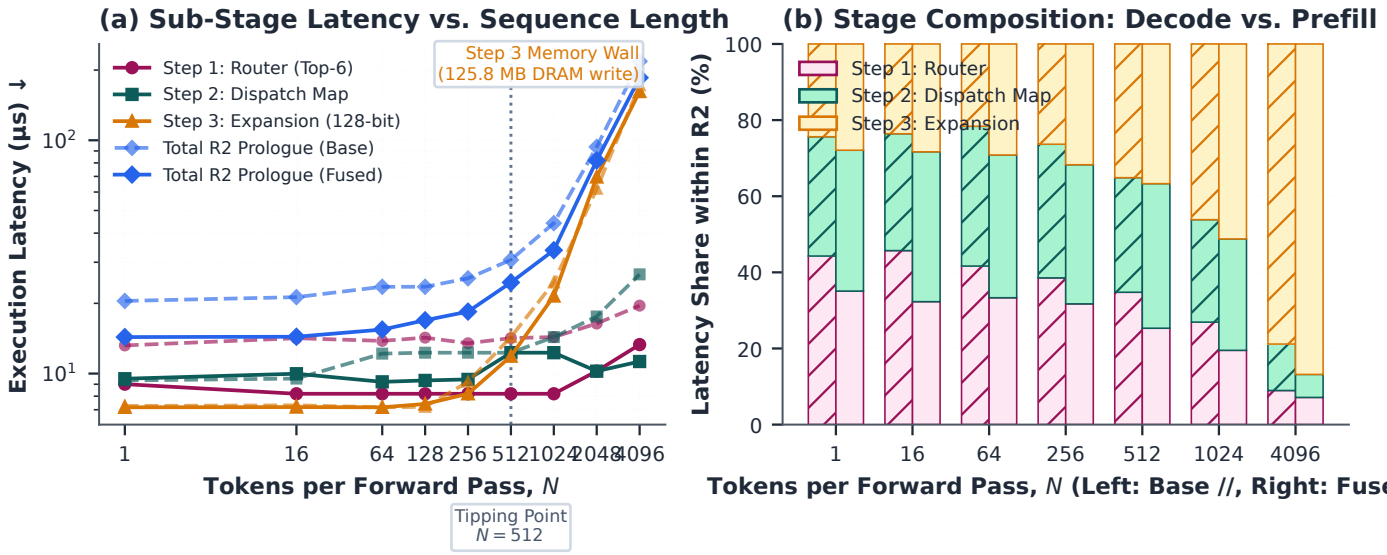
(a) Sub-Stage Latency vs. Sequence Length
(b) Stage Composition: Decode vs. Prefill
Step 3 Memory Wall
(125.8 MB DRAM write)
Step 1: Router (Top-6)
Step 2: Dispatch Map
Step 3: Expansion (128-bit)
Total R2 Prologue (Base)
Total R2 Prologue (Fused)
Execution Latency (µs) ↓
$10^2$
$10^1$
1
16
64
128
256
512
1024
2048
4096
Tokens per Forward Pass, N
Tipping Point
N = 512
Step 1: Router
Step 2: Dispatch Map
Step 3: Expansion
Latency Share within R2 (%)
100
80
60
40
20
0
1
16
64
256
512
1024
4096
Tokens per Forward Pass, N (Left: Base //, Right: Fuse